

Optimisation for Computer Vision - Report

Clément Callaert

CentraleSupélec

February 2026



CentraleSupélec

Table des matières

1	Contextualization : Where Does This Paper Fit ?	2
1.1	The Pre-existing Landscape	2
1.2	The Problem and the Novel Contribution	3
2	The Core Mechanism : SDEs, ODEs, and Proximal Operators	3
2.1	From SDE to Probability Flow ODE	3
2.2	Implicit Discretization and The Proximal Operator	3
2.3	Proximal Matching for MAP Estimation	4
3	Optimization - Vision Connection	4
4	Critical Evaluation	5
4.1	Strengths	5
4.2	Limitations	5
5	Conclusion	5
A	Appendix : Empirical Results	6
A.1	2D Datasets Generation	6
A.2	Training Stability	7
A.3	MNIST Generation and FID Evaluation	8

Résumé

Report analysis of the paper “Beyond Scores : Proximal Diffusion Models” (Fang et al., 2025), submitted for the Optimization for Computer Vision (OVO) course. The paper introduces a novel generative framework that replaces the traditional forward discretization of Stochastic Differential Equations (SDEs) with a backward, implicit discretization. By framing the reverse diffusion process as a sequence of proximal operator evaluations (seen in the course), the authors bridge continuous-time generative modeling with fundamental concepts of convex optimization.

1 Contextualization : Where Does This Paper Fit ?

1.1 The Pre-existing Landscape

Over the past few years, diffusion models have become the dominant paradigm for high-dimensional image generation. The foundational insight of these models is that one can gradually corrupt data with Gaussian noise using a forward Stochastic Differential Equation (SDE), and subsequently generate data by simulating the corresponding reverse-time SDE.

The standard approach to solving this reverse SDE is based on score matching. The reverse SDE requires knowledge of the “score function,” $\nabla \ln p_t(x)$, which is the gradient of the log-density of the perturbed data. Neural networks are trained to approximate this score by minimizing the Mean Squared Error (MSE), which mathematically corresponds to learning a Minimum Mean Square Error (MMSE) denoiser. In practice, sampling is performed using explicit numerical methods like Euler-Maruyama (a forward discretization).

1.2 The Problem and the Novel Contribution

The fundamental bottleneck of standard diffusion models is inference speed. Because explicit forward discretization schemes are highly unstable when taking large step sizes (analogous to the instability of standard gradient descent with a large learning rate), standard models require hundreds or thousands of sampling steps (Neural Function Evaluations, or NFEs) to generate a single image. The paper introduces Proximal Diffusion Models (ProxDM). The authors propose a radical shift : instead of using forward discretization and MSE-based score matching, they utilize an implicit (backward) discretization of the reverse SDE. This seemingly simple numerical choice transforms the sampling step into an optimization problem that is exactly solvable via Proximal Operators such as in the course. To achieve this, they leverage a novel “proximal matching” loss to train a neural network that acts as a Maximum A Posteriori (MAP) (comparable of a ridge regression) denoiser.

2 The Core Mechanism : SDEs, ODEs, and Proximal Operators

To understand the core contribution, we must formalize the transition from standard diffusion to proximal updates.

2.1 From SDE to Probability Flow ODE

Standard diffusion models define a forward process via an SDE. For generation, we simulate the corresponding reverse-time SDE. However, it is a well-established result that one can construct a deterministic Probability Flow ODE (PF-ODE) whose trajectories share the same marginal probability densities $p_t(x)$ as the reverse SDE. By dropping the stochastic noise term, the generative process is governed entirely by the score function $\nabla \ln p_t(x)$. A simplified step from a noisy state X_k at time t_k to a cleaner state X_{k-1} with step size γ_k can be viewed through the lens of ODE discretization.

2.2 Implicit Discretization and The Proximal Operator

The standard approach to simulating the reverse SDE is to apply a **forward (explicit) Euler-Maruyama discretization**. This method evaluates the score (the gradient of the log-density) at the current, known state X_k , and injects Gaussian noise $z_k \sim \mathcal{N}(0, I)$:

$$X_{k-1} = X_k + \gamma_k \left[\frac{1}{2} X_k + \nabla \ln p_{t_k}(X_k) \right] + \sqrt{\gamma_k} z_k \quad (1)$$

This explicit step is notoriously unstable for large step sizes γ_k , demanding a high number of iterations (NFEs) to maintain generation quality.

To remedy this, ProxDM proposes a **backward (implicit) discretization**, where the continuous-time terms are evaluated at the *unknown* destination state X_{k-1} :

$$X_{k-1} = X_k + \gamma_k \left[\frac{1}{2} X_{k-1} + \nabla \ln p_{t_{k-1}}(X_{k-1}) \right] + \sqrt{\gamma_k} z_k \quad (2)$$

Solving this implicit equation directly is intractable. However, the authors demonstrate that Equation 2 mathematically corresponds to the first-order optimality condition of a specific minimization problem. Consequently, the backward update can be elegantly rewritten using the **Proximal**

Operator of the negative log-density :

$$X_{k-1} = \text{prox}_{-\frac{2\gamma_k}{2-\gamma_k} \ln p_{t_{k-1}}} \left[\frac{2}{2-\gamma_k} (X_k + \sqrt{\gamma_k} z_k) \right] \quad (3)$$

This formulation represents a paradigm shift : the generative update intrinsically acts as a Maximum A Posteriori (MAP) denoiser applied *after* the noise injection step. By substituting the explicit score evaluation with a proximal operator, the model inherits the unconditional stability of implicit numerical methods, allowing the solver to take drastically larger step sizes without diverging.

2.3 Proximal Matching for MAP Estimation

Standard diffusion models train a denoiser D_θ using Mean Squared Error (MSE), which converges to the conditional expectation (MMSE estimate). However, the proximal operator mathematically seeks the mode (the Maximum A Posteriori, or MAP estimate). To force the network to act as a proximal operator, the authors introduce a *Proximal Matching* loss :

$$l_{PM}(x, y; \zeta) = 1 - \exp \left(-\frac{\|x - y\|_2^2}{d\zeta^2} \right) \quad (4)$$

As $\zeta \rightarrow 0$, this loss mimics an L_0 pseudo-norm. It aggressively penalizes small deviations, forcing the network to collapse its prediction to the exact mode (MAP) rather than averaging over multiple plausible clean images (MMSE).

3 Optimization - Vision Connection

This methodology is deeply intertwined with the concepts from the course, specifically Lecture III on First-order Convex Optimization.

In classical computer vision tasks (e.g., image denoising or deblurring), we often minimize an objective $f(x) + g(x)$, where $g(x)$ is a non-smooth prior like Total Variation. The **Proximal Point Algorithm** and **Forward-Backward Splitting** are used to safely "step" through these stiff optimization landscapes where explicit gradients fail or oscillate.

ProxDM explicitly transports this classical optimization paradigm into deep generative vision. In image generation, the visual quality heavily depends on the stability of the ODE solver. Explicit methods (standard diffusion) require tiny step sizes to avoid overshooting the highly non-convex data manifold, leading to hundreds of Neural Function Evaluations (NFEs). By framing generation as a sequence of proximal updates, ProxDM inherits the unconditional stability of implicit numerical methods.

From a purely visual standpoint, this connection is crucial : while the MMSE denoiser (explicit MSE) tends to produce blurry images when forced to take large steps (because it averages over uncertainty), the MAP denoiser (proximal operator) explicitly snaps to the nearest high-probability mode. This optimization choice directly solves the vision problem of generating sharp, high-frequency image features (like textures and edges) under severe computational constraints (e.g., generating an image in fewer than 10 steps).

4 Critical Evaluation

4.1 Strengths

- **Improved Inference Efficiency :** By leveraging the stability of backward discretization, ProxDM empirically demonstrates the ability to generate high-quality images in very few steps (< 10 NFE), outperforming Score ODEs and SDEs in these low-step regimes. This has massive practical implications for deploying real-time generative vision models.
- **The Hybrid Approach and Step-Size Flexibility :** The authors smartly introduce a “Hybrid” variant (PDA-hybrid) that applies a forward-backward splitting strategy at *every* timestep, rather than switching algorithms mid-generation. Specifically, it evaluates the linear drift term explicitly at the current state X_k while evaluating the score implicitly at the next state X_{k+1} . This optimization choice is highly significant because the fully backward method imposes a hard upper bound on the step size ($\gamma_k < 2$) to prevent the regularization parameter from becoming negative. By relaxing this strict constraint via the hybrid approach, the authors trade a slightly slower theoretical convergence rate for massive practical flexibility, allowing the model to generate images in very few steps.

4.2 Limitations

- **Training Complexity and the ζ Parameter :** The proximal matching loss l_{PM} relies on a hyperparameter ζ . As $\zeta \rightarrow 0$, the loss function becomes extremely sharp (mimicking L_0), which notoriously leads to vanishing gradients everywhere except very close to the target. This makes the neural network significantly harder to train than standard MSE-based models, likely requiring careful warm-ups or annealing schedules for ζ .
- **Theoretical Gap regarding Convexity :** The mathematical guarantee of the proximal operator (and its uniqueness) relies heavily on the function f being convex. However, the negative log-density of natural images (like CIFAR-10 or human faces) is highly non-convex and multimodal. While the empirical approximation works, the theoretical guarantees of proximal methods are stretched when applied to such complex distributions.
- **Limited High-Resolution Evidence :** The paper demonstrates results primarily on MNIST, CIFAR-10, and low-resolution datasets. While the theoretical NFE reduction is promising, it remains to be seen if the network can successfully learn the highly intricate proximal operators required for high-resolution, latent-space diffusion models (like Stable Diffusion).

5 Conclusion

“Beyond Scores : Proximal Diffusion Models” represents a sophisticated convergence of stochastic calculus, deep learning, and optimization theory (even physics, fokker-planck equation, heat diffusion etc). By substituting the forward-discretized score-matching paradigm with a backward-discretized proximal operator framework, it directly addresses the computational inefficiency of modern generative vision models. It is a prime example of how classical optimization principles (such as those studied in the course) continue to provide crucial breakthroughs in state-of-the-art Artificial Intelligence.

A Appendix : Empirical Results

A.1 2D Datasets Generation

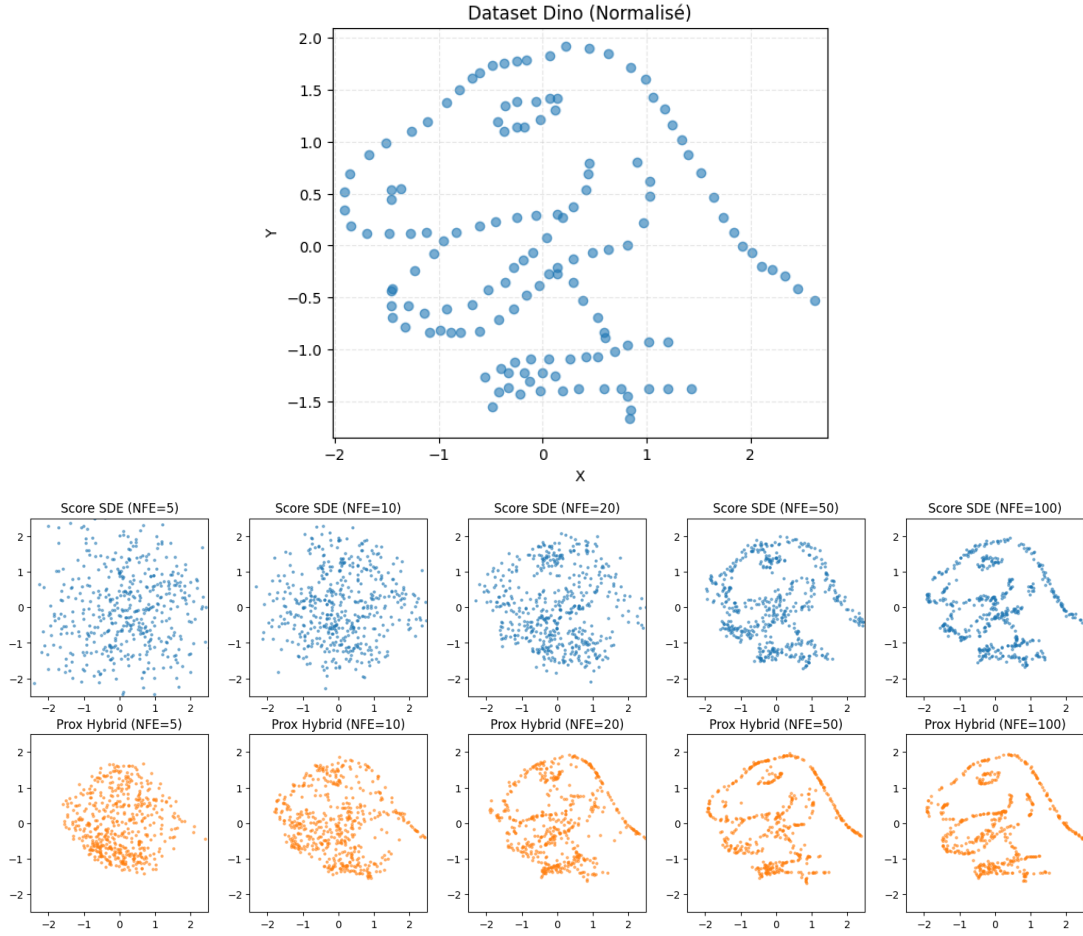


FIGURE 1 – Generation on the “Dino” dataset. ProxDM (bottom) captures the global manifold structure much faster (at 5-10 NFEs) than the standard Score SDE (top), which remains noisy.

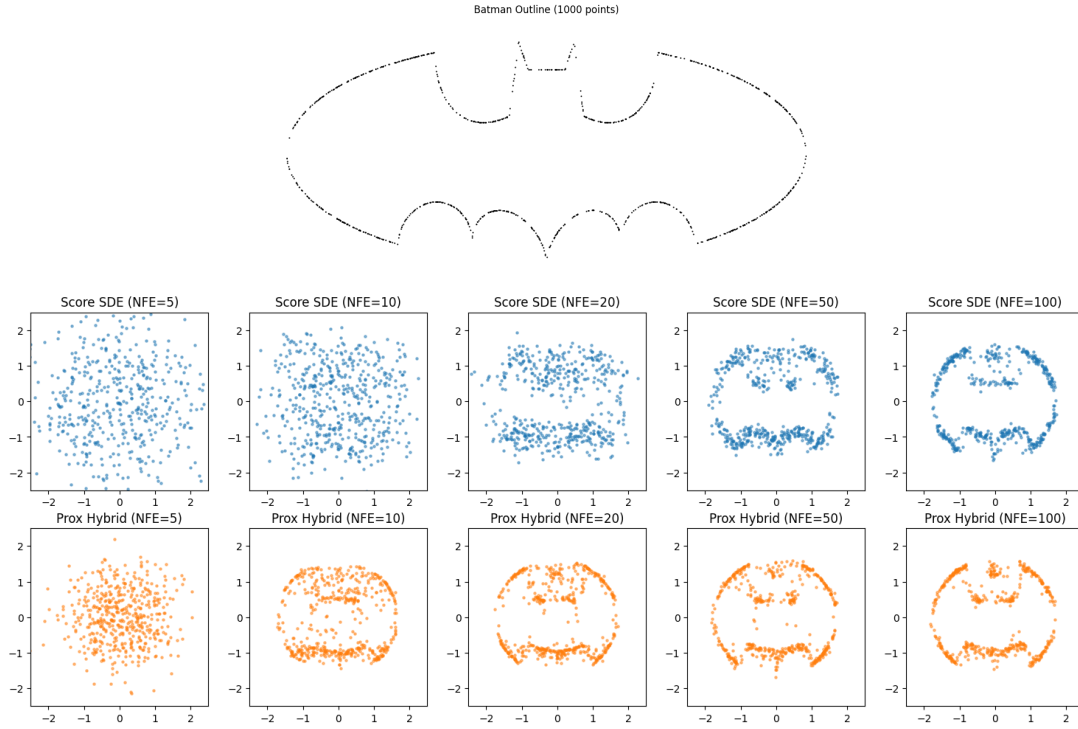


FIGURE 2 – Generation on the “Batman” dataset. Similar to the Dino dataset, the proximal backward approach demonstrates superior stability and visual convergence at very low step counts.

A.2 Training Stability

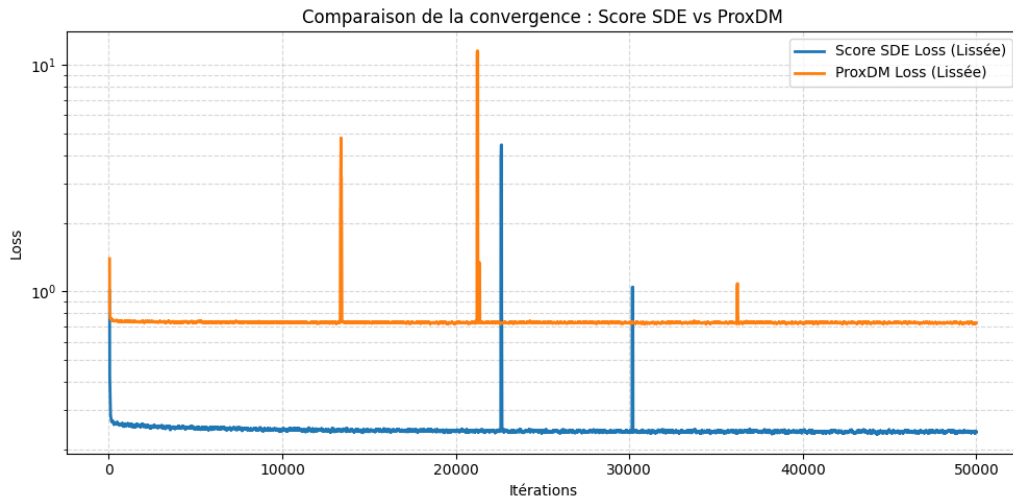


FIGURE 3 – Loss convergence comparison. The ProxDM loss exhibits significant instability spikes compared to the Score SDE, reflecting the difficulty of optimizing the sharp proximal matching objective.

A.3 MNIST Generation and FID Evaluation

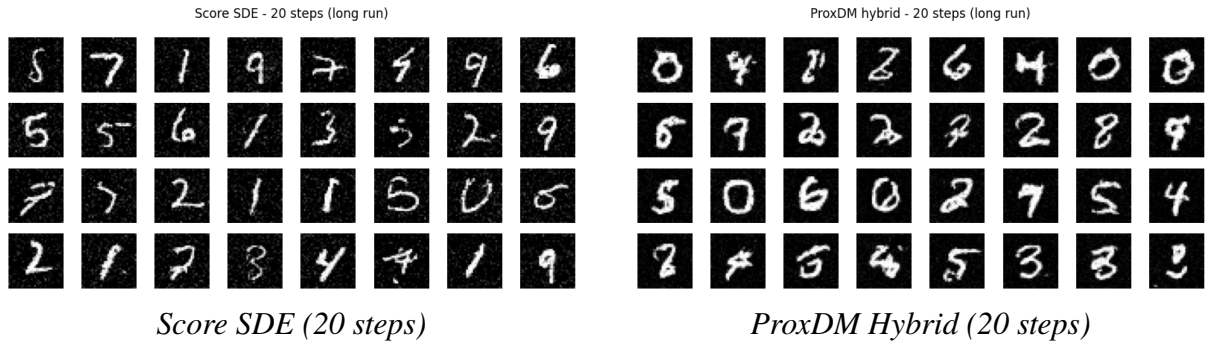


FIGURE 4 – At 20 steps, ProxDM generates visually sharper digits (acting as a MAP mode-seeker), whereas Score SDE outputs are slightly blurrier (acting as an MMSE averager).

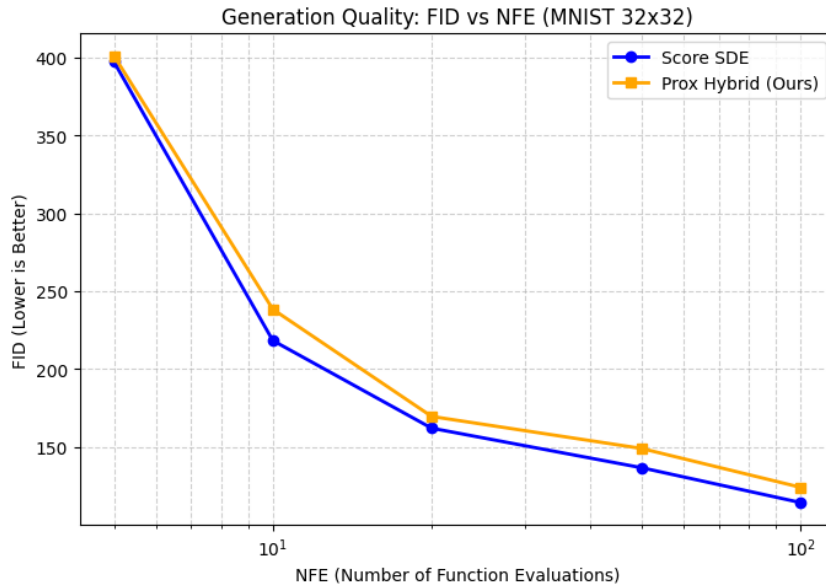


FIGURE 5 – FID vs NFE. Despite sharper individual samples, ProxDM yields a slightly worse FID score, indicating that aggressively snapping to modes may reduce the overall distributional diversity compared to Score SDE.